



XI Important Numerical Problems for Final Examination 2024

1. A watch factory claims that its watches gain or loss not more than 10 s in a year. How accurate is this watch? Express percentage.
2. If radius of a sphere is measured as 7.5 cm with error of 0.03 cm, find the approximate error in calculating its volume.
3. A helicopter is ascending at a rate of 12 m/s. At a height of 80 m above the ground, a package is dropped. How long does the package take to reach the ground?
4. Two tug boats are towing a ship each exert a force of 6000 N, and the angle between two ropes is 60°. Calculate the resultant force on the ship.
5. At what angle range of projectile becomes equal to the maximum height?
6. A 50 g bullet is fired into a 10 kg block that is suspended by a long cord so that it can swing as a pendulum. Due to the impact center of gravity of block rises by 10 cm, what was the speed of bullet?
7. A truck weighing 2500 kg and moving with a velocity of 21 m/s collides with a stationary car weighing 1000 kg. The truck and car move together after the impact. Calculate their common velocity.
8. What is the angular momentum of a 2.9 kg uniform cylindrical grinding wheel of radius 20 cm when rotating at 1550 rpm? How much torque is required to stop it in 6 s?
9. A thin disk with a 0.3 m diameter and a total moment of inertia of 0.45 kg.m² is rotating about its center of mass. There are three rocks with masses of 0.2 kg each on the outer part of the disk. Find the total moment of inertia of the system.
10. A satellite wishes to orbit the earth at a height of 100 km above the Earth's surface. Determine the speed, acceleration and orbital period of the satellite.
11. When an object is thrown upwards it rises to a height h, how high is the object in terms of h when it has lost one third of its original kinetic energy?
12. A motor pumps the water at a rate of 500 g/min to a height of 120 m. If the motor is 50% efficient then how much input electric power is needed?
13. The pressure in a water pipe in the ground floor of a building is 4×10^5 Pa but three floors up it is only 2×10^5 Pa. What is the height between the ground floor and the third floor? Assume the water to be stationary.
14. A spherical balloon has a radius of 7.35 m and is filled with helium (0.179 kg/m^3). How large a cargo can it lift, assuming that the skin and structure of the balloon have a mass of 930 kg. Neglect the buoyant force on the cargo volume itself.
15. Calculate the viscous drag on a drop of oil of 0.1 mm radius falling through air at the terminal velocity. (viscosity of air = 1.8×10^{-5} Pa.s, density of oil = 850 kg/m^3)
16. What is the maximum weight of an aircraft with a wing area of 50 m² flying horizontally, the velocity of air over the upper surface of the wing is 150 m/s and that the lower surface is 140 m/s. Density of air is 1.29 kg/m^3 .
17. Water circulates throughout a house in a hot water heating system. If water is pumped at a speed of 0.50 m/s through a 4.0 cm diameter pipe in the basement under a pressure of 3.0 atm, what will be the flow speed and pressure in a 2.6 cm diameter pipe on the second floor 5.0 m above? Assume the pipes do not divide into branches.
18. Find the intensity of electric field at a point such that a proton placed at it would experience a force equal to its weight.



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8. 19. Three charges $q_1=2 \mu\text{C}$, $q_2=3 \mu\text{C}$, and $q_3=4 \mu\text{C}$ are placed in air at the vertices of an equilateral triangle of sides 10 cm. calculate the magnitude of the resultant force on charge q_3 .
8. 20. An electron of charge $1.6 \times 10^{-19} \text{ C}$ is situated in a uniform electric field of intensity $1200 \text{ V}\cdot\text{cm}^{-1}$. Find the force on it, its acceleration, and the time it takes to travel 2 cm from rest. $m=9.11 \times 10^{-31} \text{ kg}$ $t = \text{formula} = ?$
9. 21. What capacitance is required to store energy of 10 kWh at a potential difference of 1000 V?
9. 22. Capacitor is charged through a large non-reactive resistance by a battery of constant voltage V . For this arrangement, if the capacitor has a capacitance of $10 \mu\text{F}$ and the resistance is $10 \text{ M}\Omega$, calculate the time taken for the capacitor to receive 90% of its final charge.
10. 23. A negligible small current is passed through a wire of length 12 m and uniform cross section $4.0 \times 10^{-7} \text{ m}^2$, and its resistance is measured to be 6.0Ω . What is the resistivity of the material at the temperature of the experiment?
10. 24. A given battery has a 12 V emf and an internal resistance of 0.1Ω . Calculate its terminal voltage when connected to a 10Ω load. If the internal resistance increases to 0.5Ω , find terminal voltage and the current.
11. 25. The amplitude of oscillation of a pendulum decays by a factor of 20 in 120 s. By what factor has its energy decayed in that time?
11. 26. A sewing machine needle moves with a rapid vibratory motion like SHM, as it sews a seam. Suppose the needle moves 8.4 mm from its highest to its lowest position and it make 24 stitches in 9 s. What is the maximum needle speed?
11. 27. A pendulum of length 75 cm and mass 2.5 kg swings with a mechanical energy of 0.015 J. What is its amplitude?
12. 28. A train sounds its whistle while passing by a railroad crossing. An observer at the crossing measures a frequency of 219 Hz as the train approaches the crossing and a frequency of 184 Hz as the train leaves. The speed of sound is 340 m/s. Find the speed of the train and the frequency of its whistle.
12. 29. In order to decrease the fundamental frequency of a guitar string by 4.0 %, by what percentage should you reduce the tension?
12. 30. If the velocity of sound in air at 27°C and at a pressure of 76 cm of Hg is 345 m/s. Find the velocity at 127°C and 75 cm of Hg
13. 31. A grating has exactly 8000 lined uniformly spaced over 2.54 cm and is illuminated by light from a mercury vapor discharge lamp. What is the expected angle for the third order maximum of the light of wavelength 546 nm?
13. 32. A beam of X-rays of wavelength 0.071 nm is diffracted by a plane of rock salt with a distance between the atomic planes are $1.98 \text{ \AA}$. Find the glancing angle for the second order diffraction.
13. 33. A monochromatic light of wavelength $6900 \text{ \AA}$ is used to illuminate two parallel slits. On a screen that is 3.30 m away from the slits, interference fringes are observed. The distance between adjacent bright fringes in the center of pattern is 1.80 cm, what is the distance between the slits?

Numerical, no. 1;

Data;

$$t = 10 \text{ sec}$$

$$\% = ?$$

Solution;

$$\bullet \% = \frac{t}{\text{total sec in a year}}$$

$$\% = \frac{10}{365 \times 24 \times 60 \times 60}$$

$$\% = 3.17 \times 10^{-7} \% \quad \underline{\text{Ans!}}$$

Numerical no. 2;

Data;

$$r = 7.5 \text{ cm}$$

$$\Delta r = 0.03 \text{ cm}$$

$$\frac{\Delta V}{V} = ?$$

$$V = ?$$

Solution;

$$\frac{\Delta V}{V} = 3 \frac{\Delta r}{r}$$

$$\frac{\Delta V}{V} = 3 \frac{0.03}{7.5}$$

$$\boxed{\frac{\Delta V}{V} = 0.012} \text{ Ans!}$$

volume of sphere;

$$V = \frac{4}{3} \pi r^3$$

$$V = \frac{4}{3} (3.142) (7.5)^3$$

$$\boxed{V = 1767.375 \text{ cm}^3} \text{ Ans!}$$

Numerical no. 3;

Data;

$$v = 12 \text{ m/s}$$

$$h = 80 \text{ m}$$

$$t = ?$$

Solution;

$$s = v_i t + \frac{1}{2} a t^2$$

$$h = v_i t + \frac{1}{2} g t^2$$

$$-80 = 12t + \frac{1}{2} (9.8) t^2$$

$$-80 = 12t + 4.9 t^2$$

$$-80 - 12t - 4.9 t^2 = 0$$

$$\frac{-4.9 t^2}{a} - \frac{12t}{b} - \frac{80}{c} = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-(-12) \pm \sqrt{(-12)^2 - 4(-4.9)(-80)}}{2(-4.9)}$$

$$t = 5.4 \text{ sec} \quad \text{Ans!}$$

numerical no. 4;

Data;

$$F_1 = 6000 \text{ N}$$

$$F_2 = 6000 \text{ N}$$

$$\theta = 60^\circ$$

$$R_F = ?$$

Solution;

$$R_F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$

$$R_F = \sqrt{6000^2 + 6000^2 + 2(6000)(6000)\cos 60}$$

$$R_F = 10392.304 \text{ N} \quad \text{Ans!}$$

Numerical no. 5;

Data;

$$R = H$$

$$\theta = ?$$

$$R = \frac{V_0^2 \sin 2\theta}{g}$$

$$H = \frac{V_0^2 \sin^2 \theta}{2g}$$

Solution;

$$\frac{V_0^2 \sin 2\theta}{g} = \frac{V_0^2 \sin^2 \theta}{2g} \quad \therefore \sin 2\theta = 2 \sin \theta \cos \theta$$

$$2 \sin \theta \cos \theta = \frac{\sin^2 \theta}{2}$$

$$4 \cos \theta = \sin \theta$$

$$4 = \frac{\sin \theta}{\cos \theta}$$

$$\therefore \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$4 = \tan \theta$$

$$\theta = \tan^{-1}(4)$$

$$\theta = 76^\circ \text{ Ans!}$$

Numerical no. 6;

Data;

$$m_1 = 50 \text{ g} = \frac{50}{1000} = 0.05 \text{ kg}$$

$$m_2 = 10 \text{ kg}$$

$$h = 10 \text{ cm} = \frac{10}{100} = 0.1 \text{ m}$$

$$u_2 = 0 \text{ m/s}$$

$$v_1 = v_2 = v = ?$$

Solution;

$$\bullet m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$0.05 \times u_1 + (10 \times 0) = v (m_1 + m_2)$$

$$0.05 u_1 = v (0.05 + 10)$$

$$0.05 u_1 = v \cdot 10.05$$

$$u_1 = \frac{10.05}{0.05}$$

$$u_1 = 201 \text{ m/s}$$

$$\bullet \text{loss in K.E} = \text{Gain in P.E}$$

$$\frac{1}{2} m v^2 = mgh$$

$$v = \sqrt{2gh}$$

$$v = \sqrt{2(9.8)(0.1)}$$

$$v = 1.4 \text{ m/s}$$

$$\bullet u_1 = 201 \times 1.4$$

$$\boxed{u_1 = 281.4 \text{ m/s}} \text{ Ans!}$$

Numerical no. 7;

Data;

$$m_1 = 2500 \text{ kg}$$

$$u_1 = 21 \text{ m/s}$$

$$m_2 = 1000 \text{ kg}$$

$$u_2 = 0 \text{ m/s}$$

$$v_1 = v_2 = v = ?$$

Solution;

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$(2500 \times 21) + (1000 \times 0) = v(2500 + 1000)$$

$$52500 = v(3500)$$

$$v = \frac{52500}{3500}$$

$$\boxed{v = 15 \text{ m/s} \text{ (Ans)}}$$

Numerical no. 8,

Data;

$$m = 2.9 \text{ kg}$$

$$r = 20 \text{ cm} = \frac{20}{100} = 0.2 \text{ m}$$

$$\text{rpm} = 1550$$

$$t = 6 \text{ s}$$

$$L = ?$$

$$T = ?$$

Solution;

$$\bullet I = \frac{1}{2} m r^2$$

$$I = \frac{1}{2} (2.9) (0.2)^2$$

$$I = 0.058 \text{ kg} \cdot \text{m}^2$$

$$\bullet \omega = \frac{2\pi \text{ rpm}}{60}$$

$$\omega = \frac{2(3.142)(1550)}{60}$$

$$\omega = 162.33 \text{ rad/sec}$$

$$\bullet L = I \cdot \omega$$

$$L = 0.058 (162.33)$$

$$\boxed{L = 9.415 \text{ kg} \cdot \text{m}^2/\text{sec}} \text{ Ans!}$$

$$\bullet T = \frac{L}{\Delta t}$$

$$T = \frac{9.415}{6}$$

$$\boxed{T = 1.569 \text{ N} \cdot \text{m}^2} \text{ Ans!}$$

Numerical no. 9;

Data;

$$d = 0.3\text{m} = r = \frac{d}{2} = \frac{0.3}{2} = 0.15\text{m}$$

$$I = 0.45\text{ kg}\cdot\text{m}^2$$

$$m_1 = m_2 = m_3 = m = 0.2\text{ kg}$$

$$I_s = ?$$

Solution;

$$I_s = I + \sum m r^2$$

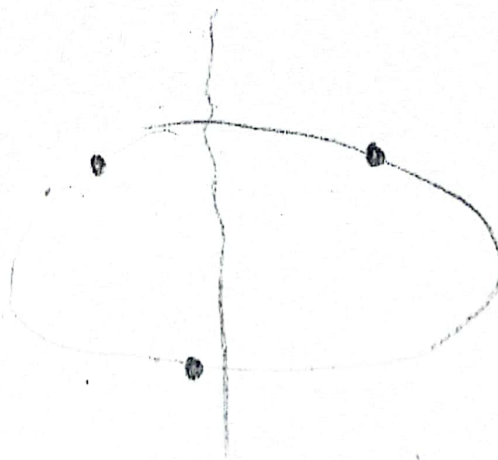
$$I_s = I + m_1 r^2 + m_2 r^2 + m_3 r^2$$

$$= m_1 = m_2 = m_3 = m$$

$$I_s = I + 3m r^2$$

$$I_s = 0.45 + 3(0.2)(0.15)^2$$

$$I_s = 0.463\text{ kg}\cdot\text{m}^2 \quad \underline{\text{Ans!}}$$



Numerical no. 10;

Data;

$$h = 100 \text{ km} = \frac{100}{1000} = 0.1 \text{ m}$$

1) $V = ?$

2) $a = ?$

3) $T = ?$

Solution;

$$\therefore R_E = 6.37 \times 10^6 \text{ m}$$

$$R = h + R_E$$

$$R = 0.1 + 6.37 \times 10^6$$

$$R = 6.37 \times 10^6 \text{ m}$$

$$1) V = \sqrt{\frac{GM_E}{R}}$$

$$\therefore G = 6.67 \times 10^{-11}$$

$$\therefore M_E = 5.98 \times 10^{24}$$

$$1) V = \sqrt{\frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{6.37 \times 10^6}}$$

$$V = 7.913 \times 10^3 \text{ m/s} \quad \underline{\text{Ans!}}$$

$$2) a = \frac{v^2}{R}$$

$$a = \frac{(7.913 \times 10^3)^2}{6.37 \times 10^6}$$

$$a = 9.829 \text{ m/s}^2 \quad \underline{\text{Ans!}}$$

$$3) T^2 = \frac{4\pi^2 R^3}{GM_E}$$

$$T = \sqrt{\frac{4\pi^2 R^3}{GM_E}}$$

$$T = \sqrt{\frac{4(3.142)^2 (6.37 \times 10^6)^3}{6.67 \times 10^{-11} (5.98 \times 10^{24})}}$$

$$T = 5.058 \times 10^3 \text{ sec}$$

$\therefore T$ in hours;

$$T = \frac{5.058 \times 10^3}{60 \times 60}$$

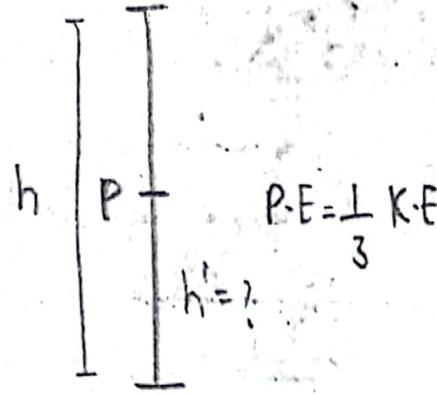
$$T = 1.405 \text{ hours} \quad \underline{\text{Ans!}}$$

Numerical no. 11;

Data;

$$h = \frac{1}{3} K \cdot E$$

$$h' = ?$$



Solution;

- At the top;

loss of $K.E = \text{Gain in } P.E$

$$\frac{1}{2} mv^2 = mgh$$

- At point P ,

$$P.E = \frac{1}{3} K.E \quad \therefore K.E = mgh$$

$$mgh' = \frac{1}{3} (mgh)$$

$$h' = \frac{1}{3} h$$

$$\boxed{h' = \frac{h}{3}} \quad \text{Ans!}$$

Numerical no. 12;

Data;

$$h = 120 \text{ m}$$

$$\eta = 50\%$$

$$m = 500 \text{ g} = \frac{500}{1000} = 0.5 \text{ kg}$$

$$t = 1 \text{ min} = 60 \text{ sec}$$

$$P_{\text{input}} = ?$$

Solution;

$$P_{\text{output}} = \frac{mgh}{t}$$

$$P_{\text{output}} = \frac{0.5 (9.8) (120)}{60}$$

$$P_{\text{output}} = 9.8 \text{ W}$$

$$\eta = \frac{P_{\text{output}}}{P_{\text{input}}} \times 100$$

$$P_{\text{input}} = \frac{P_{\text{output}}}{\eta} \times 100$$

$$P_{\text{input}} = \frac{9.8}{50} \times 100$$

$$P_{\text{input}} = 19.6 \text{ W} \quad \underline{\text{Ans!}}$$

Numerical no. 13;

Data;

$$P_1 = 4 \times 10^5 \text{ Pa}$$

$$P_2 = 2 \times 10^5 \text{ Pa}$$

$$h = ? \quad \rho_{\text{water}} = 1000 \text{ kg/m}^3$$

Solution;

$$P_1 - P_2 = \rho gh$$

$$4 \times 10^5 - 2 \times 10^5 = 1000 (9.8) h$$

$$2 \times 10^5 = 9800 h$$

$$h = \frac{2 \times 10^5}{9800}$$

$$h = 20.408 \text{ m} \quad \underline{\text{Ans!}}$$

Numerical no. 14;

Data;

$$r = 7.35 \text{ m}$$

$$m_B = 930 \text{ kg}$$

$$m_C = ?$$

Solution;

$$V = \frac{4}{3} \pi r^3$$

$$V = \frac{4}{3} (3.142) (7.35)^3$$

$$V = 1.663 \times 10^3 \text{ m}^3$$

$$F = \rho_{\text{air}} g V$$

$$\therefore \rho_{\text{air}} = 1.29 \text{ kg/m}^3$$

$$F = (1.29)(9.8)(1.663 \times 10^3)$$

$$F = 2.102 \times 10^4 \text{ N}$$

$$m_{\text{He}} = \rho_{\text{He}} V$$

$$\therefore \rho_{\text{He}} = 0.179 \text{ kg/m}^3$$

$$m_{\text{He}} = (0.179)(1.663 \times 10^3)$$

$$m_{\text{He}} = 2.976 \times 10^2 \text{ kg}$$

$$F = m_B g + m_{\text{He}} g + m_C g$$

$$F = g (m_B + m_{\text{He}} + m_C)$$

$$\frac{F}{g} = m_B + m_{\text{He}} + m_C$$

$$\frac{2.102 \times 10^4}{9.8} = 930 + 2.976 \times 10^2 + m_C$$

$$2.144 \times 10^3 = 1.227 \times 10^3 + m_C$$

$$2.144 \times 10^3 - 1.227 \times 10^3 = m_C$$

$$m_C = 2.021 \times 10^3 \text{ kg} \text{ Ans!}$$

Numeical no. 15;

Data;

$$r = 0.1 \text{ mm} = 0.1 \times 10^{-3} \text{ m}$$

$$\eta_{\text{air}} = 1.8 \times 10^{-5} \text{ Pa/sec OR } \eta$$

$$\rho_{\text{oil}} = 850 \text{ kg/m}^3$$

$$F = ?$$

Solution;

$$v_t = \frac{2}{9} \times \frac{\rho_{\text{oil}}}{\eta} r^2 g$$

$$v_t = \frac{2}{9} \times \frac{850}{1.8 \times 10^{-5}} \times (0.1 \times 10^{-3})^2 \times (9.8)$$

$$v_t = 1.028 \times 10^{-12} \text{ m/s}$$

$$F = 6 \pi \eta r v_t$$

$$F = 6(3.142)(1.8 \times 10^{-5})(0.1 \times 10^{-3})(1.028 \times 10^{-12})$$

$$\boxed{F = 3.488 \times 10^{-10} \text{ N}} \text{ Ans!}$$

Numerical no. 16;

Data;

$$A_1 = 50 \text{ m}^2$$

$$V_1 = 150 \text{ m/s}$$

$$V_2 = 140 \text{ m/s}$$

$$\rho_{\text{air}} = 1.29 \text{ kg/m}^3$$

$$W = ?$$

Solution;

$$P_1 + \frac{1}{2} \rho V_1^2 + \rho gh = P_2 + \frac{1}{2} \rho V_2^2 + \rho gh$$

$$P_2 - P_1 = -\frac{1}{2} \rho V_2^2 + \frac{1}{2} \rho V_1^2$$

$$P_2 - P_1 = -\frac{1}{2} (1.29) (140)^2 + \frac{1}{2} (1.29) (150)^2$$

$$P_2 - P_1 = 1870.5 \text{ Pa}$$

$$\therefore P_2 - P_1 = \Delta P$$

$$W = \Delta P \times A$$

$$W = 1870.5 \times 50$$

$$W = 9.35 \times 10^4 \text{ N} \quad \underline{\text{Ans!}}$$

Numerical no. 17;

$$1 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$$

Data;

$$V_1 = 0.50 \text{ m/s}$$

$$d_1 = 4.0 \text{ cm} = 0.04 \text{ m}$$

$$h_1 = 0 \text{ m}$$

$$P_1 = 3.0 \text{ atm} = 3.0(1.01 \times 10^5) = 3.03 \times 10^5 \text{ Pa}$$

$$d_2 = 2.6 \text{ cm} = 0.026 \text{ m}$$

$$h_2 = 5.0 \text{ m}$$

$$V_2 = ?$$

$$P_2 = ?$$

$$\therefore \rho_{\text{water}} = 1000 \text{ kg/m}^3$$

Solution;

$$A_1 = \frac{\pi d_1^2}{4}$$

$$A_1 = \frac{(3.14)(0.04)^2}{4}$$

$$A_1 = 1.256 \times 10^{-3} \text{ m}^2$$

$$A_2 = \frac{\pi d_2^2}{4}$$

$$A_2 = \frac{(3.14)(0.026)^2}{4}$$

$$A_2 = 5.309 \times 10^{-4} \text{ m}^2$$

$$A_1 V_1 = A_2 V_2$$

$$V_2 = \frac{A_1 V_1}{A_2}$$

$$V_2 = \frac{(1.256 \times 10^{-3})(0.50)}{(5.309 \times 10^{-4})}$$

$$V_2 = 1.179 \text{ m/s} \quad \underline{\text{Ans!}}$$

$$P_1 + \frac{1}{2} \rho V_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho V_2^2 + \rho g h_2$$

$$P_2 = P_1 + \frac{1}{2} \rho V_1^2 + \rho g h_1 - \frac{1}{2} \rho V_2^2 - \rho g h_2$$

$$P_2 = 3.03 \times 10^5 + \frac{1}{2} (1000)(0.50)^2 - \frac{1}{2} (1000)(1.179)^2 - (1000)(9.8)(5.0)$$

$$P_2 = -2.534 \times 10^5 \text{ Pa} \quad \underline{\text{Ans!}}$$

Numeical no. 18;

Data;

$$m_p = 1.672 \times 10^{-27} \text{ kg}$$

$$q_p = 1.602 \times 10^{-19} \text{ C}$$

$$F = W = mg$$

$$E' = ?$$

Solution;

$$E = \frac{F}{q}$$

$$E = \frac{m_p g}{q_p}$$

$$E = \frac{(1.672 \times 10^{-27})(9.8)}{(1.602 \times 10^{-19})}$$

$$E = 1.022 \times 10^{-7} \text{ N/C} \text{ Ans!}$$

Numerical no. 19;

Data;

$$q_1 = 2 \mu C = 2 \times 10^{-6} C$$

$$q_2 = 3 \mu C = 3 \times 10^{-6} C$$

$$q_3 = 4 \mu C = 4 \times 10^{-6} C$$

$$Y = 10 \text{ cm} = \frac{10}{100} = 0.1 \text{ m}$$

$$F_R = ? \quad ; K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

Solution;

$$F_1 = \frac{K q_1 q_3}{r^2}$$

$$F_1 = \frac{9 \times 10^9 (2 \times 10^{-6}) (4 \times 10^{-6})}{(0.1)^2}$$

$$F_1 = 7.2 \text{ N}$$

$$F_2 = \frac{K q_2 q_3}{r^2}$$

$$F_2 = \frac{9 \times 10^9 (3 \times 10^{-6}) (4 \times 10^{-6})}{(0.1)^2}$$

$$F_2 = 10.8 \text{ N}$$

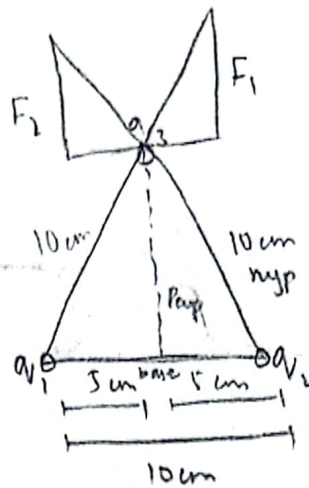
$$\cos \theta = \frac{B}{H}$$

$$\cos \theta = \frac{5}{10}$$

$$\cos \theta = 0.5$$

$$\theta = \cos^{-1}(0.5)$$

$$\theta = 60^\circ$$



$$F_R = \sqrt{F_1^2 + F_2^2 + 2 F_1 F_2 \cos \theta}$$

$$F_R = \sqrt{(7.2)^2 + (10.8)^2 + 2(7.2)(10.8) \cos(60^\circ)}$$

$$F_R = 15.69 \text{ N} \quad \underline{\text{Ans!}}$$

N Numerical no. 20;

1 Data;

0 $q_e = 1.6 \times 10^{-19} \text{ C}$

0 $E = 1200 \text{ V-cm}^{-1}$

0 $h = 2 \text{ cm} = \frac{2}{100} = 0.02 \text{ m}$

$m_e = 9.11 \times 10^{-31} \text{ kg}$

1) $F = ?$

2) $a = ?$

3) $t = ?$

• F Solution;

• $F = E q_e$

F $F = 1200 (1.6 \times 10^{-19})$

$F = 1.92 \times 10^{-16} \text{ N}$ Ans!

F $a = \frac{F}{m}$

• F $a = \frac{1.92 \times 10^{-16}}{9.11 \times 10^{-31}}$

F $a = 2.10 \times 10^{14} \text{ m/s}^2$ Ans!

• For t ;

F $S = Vit + \frac{1}{2} at^2$

$\therefore V = 0 \text{ m/s (rest)}$

• $0.02 = \frac{1}{2} (2.0 \times 10^{14}) t^2$

$0.02 = 1 \times 10^{14} t^2$

$\frac{0.02}{1 \times 10^{14}} = t^2$

$2 \times 10^{-16} = t^2$

sq on both sides

$\sqrt{2 \times 10^{-16}} = \sqrt{t^2}$

$t = 1.414 \times 10^{-8} \text{ sec}$

Numerical no. 21;

Data;

$$U = 10 \text{ kWh} = 10 \times 1000 \times 3600 = 3.6 \times 10^7 \text{ J}$$

$$V = 1000 \text{ V}$$

$$C = ?$$

Solution;

$$U = \frac{1}{2} CV^2$$

$$C = \frac{2U}{V^2}$$

$$C = \frac{2(3.6 \times 10^7)}{(1000)^2}$$

$$C = 72 \text{ F} \quad \underline{\text{Ans!}}$$

Numerical no. 22;

Data;

$$C = 10 \mu F = 10 \times 10^{-6} F$$

$$R = 10 M \Omega = 1 \times 10^6 \Omega$$

$$Q = 90\% \quad \therefore Q_0 = 0.90 Q_0$$

$$t = ?$$

Solution;

$$Q = Q_0 (1 - e^{-\frac{t}{RC}})$$

$$0.90 Q_0 = Q_0 (1 - e^{-\frac{t}{RC}})$$

$$0.90 = 1 - e^{-\frac{t}{RC}}$$

$$e^{-\frac{t}{RC}} = 1 - 0.90$$

$$e^{-\frac{t}{RC}} = 0.1$$

Applying $\ln$ on both sides;

$$\ln e^{-\frac{t}{RC}} = \ln(0.1) \quad \therefore \ln e = 1$$

$$\frac{-t}{RC} = -2.30$$

$$t = 2.30(RC)$$

$$t = 2.30(1 \times 10^6)(10 \times 10^{-6})$$

$$\boxed{t = 23 \text{ sec}} \quad \underline{\text{Ans!}}$$

"above tan we have $\ln$ button in scientific

Numerical no. 23;

Data;

$$L = 12 \text{ m}$$

$$A = 4.0 \times 10^{-7} \text{ m}^2$$

$$R = 6.0 \, \Omega$$

$$\rho = ?$$

Solution;

$$R = \frac{\rho L}{A}$$

$$\rho = \frac{RA}{L}$$

$$\rho = \frac{6.0 (4.0 \times 10^{-7})}{12}$$

$$\rho = 2 \times 10^{-7} \, \Omega \text{ m} \quad \underline{\underline{\text{Ans!}}}$$

Numerical no. 24;

Data;

$$\mathcal{E} = 12 \text{ V}$$

$$r = 0.1 \Omega$$

$$R = 10 \Omega$$

$$r' = 0.5 \Omega$$

$$V_t = ?$$

$$I = ?$$

= jb question m internal resistance
k sth load attach hoga tu
discharging ka case hoga otherwise
charging ka.

Solution;

For $r = 0.1 \Omega$

$$I = \frac{\mathcal{E}}{R + r}$$

$$I = \frac{12}{10 + 0.1}$$

$$I = 1.18 \text{ Amp} \quad \underline{\text{Ans!}}$$

$$V_t = \mathcal{E} - Ir$$

$$V_t = 12 - 1.18(0.1)$$

$$V_t = 11.882 \text{ volts} \quad \underline{\text{Ans!}}$$

For $r' = 0.5 \Omega$

$$I = \frac{\mathcal{E}}{R + r'}$$

$$I = \frac{12}{10 + 0.5}$$

$$I = 1.14 \text{ Amp} \quad \underline{\text{Ans!}}$$

$$V_t = \mathcal{E} - Ir'$$

$$V_t = 12 - 1.14(0.5)$$

$$V_t = 11.43 \text{ volts} \quad \underline{\text{Ans!}}$$

Numerical no. 25,

Data;

$$u_0 = \frac{u_0}{20}$$

$$t = 120 \text{ sec}$$

$$E' = ?$$

Solution;

$$\therefore K \cdot E = \frac{1}{2} m v^2 \quad \therefore m = k \quad \therefore v = u_0$$

$$E' = \frac{1}{2} k u_0^2$$

$$E' = \frac{1}{2} k \left(\frac{u_0}{20} \right)^2$$

$$E' = \frac{1}{2} k \frac{u_0^2}{400}$$

$$E' = \frac{E}{400} \quad \underline{\underline{\text{Ans!}}}$$

Numeical no. 26,

Data;

$$2u_0 = 8.4 \text{ mm} = \frac{8.4}{2} = 4.2 \times 10^{-3} \text{ m}$$

$$n = 24$$

$$t = 9 \text{ sec}$$

$$v = ?$$

Solution;

$$\bullet f = \frac{n}{t} = \frac{24}{9}$$

$$f = 2.66 \text{ Hz}$$

$$\bullet \omega = 2\pi f$$

$$\omega = 2(3.142)(2.66)$$

$$\omega = 16.71 \text{ rad/sec}$$

$$\bullet v = \omega u_0$$

$$v = 16.71(4.2 \times 10^{-3})$$

$$\boxed{v = 0.070 \text{ m/s}} \text{ Ans!}$$

Numerical no. 27,

Data;

$$L = 75 \text{ cm} = \frac{75}{100} = 0.75 \text{ m}$$

$$m = 2.5 \text{ kg}$$

$$ME = 0.015 \text{ J}$$

$$u_0 = ?$$

Solution;

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$T = 2(3.142) \sqrt{\frac{0.75}{9.8}}$$

$$T = 1.738 \text{ sec}$$

$$\omega = \frac{2\pi}{T}$$

$$\omega = \frac{2(3.142)}{1.738}$$

$$\omega = 3.615 \text{ rad/sec}$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$3.615 = \sqrt{\frac{k}{2.5}}$$

sq on both sides;

$$(3.615)^2 = \left(\sqrt{\frac{k}{2.5}}\right)^2$$

$$13.068 = \frac{k}{2.5}$$

$$k = 13.068 \times 2.5$$

$$k = 32.67 \text{ N/m.}$$

$$E = \frac{1}{2} k u_0^2$$

$$0.015 = \frac{1}{2} (32.67) u_0^2$$

$$\frac{0.015 \times 2}{32.67} = u_0^2$$

$$u_0^2 = 9.182$$

sq on both sides

$$\sqrt{u_0^2} = \sqrt{9.182}$$

$$u_0 = 3.030 \text{ m} \text{ Ans!}$$